

# Cell Cycle: Copy, Check, Divide

**I can describe the cell cycle, distinguishing interphase from mitosis and cytokinesis, and sequence prophase, metaphase, anaphase and telophase using chromosome evidence.**

## **COPY DNA**

during interphase

## **SEPARATE COPIES**

during mitosis

## **DIVIDE CELL**

cytokinesis

[Open the original interactive lesson](#)

## What success looks like

- Describe interphase as growth, increase of subcellular structures and DNA replication before mitosis.
- Sequence prophase, metaphase, anaphase and telophase using the positions and appearance of chromosomes.
- Explain that cytokinesis divides the cytoplasm and membrane after nuclear division to form two daughter cells.

## Retrieve and reconnect

- State where chromosomes are found in an animal or plant cell.
- Explain why DNA must be copied before one cell forms two cells.
- Distinguish growth of one cell from an increase in cell number.
- Lead-in: predict what must happen to chromosome copies before cytokinesis.

**A cell has grown and copied its DNA, but the chromosome copies have not yet separated. Which part of the cycle is it completing?**

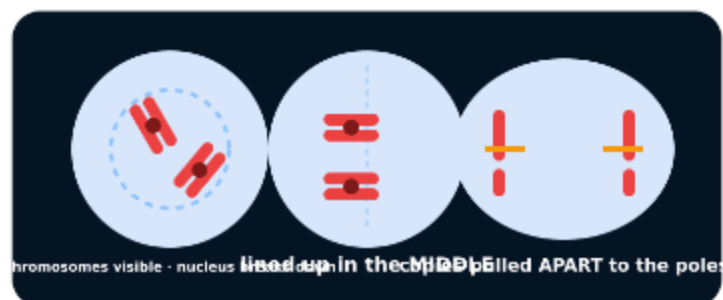
Think first. Point, say, sign or write your choice. Give one reason.

- A. Interphase, which prepares the cell for nuclear division.
- B. Cytokinesis, because the cytoplasm has already formed two cells.
- C. The end of the cycle, because DNA copying alone makes two daughter cells.

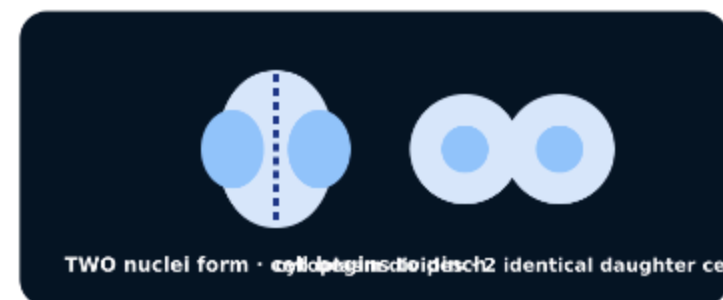
# Look closely · what is the evidence?



Interphase cell with intact nucleus



Prophase, metaphase, anaphase



Telophase then cytokinesis

## Words we will use

- interphase — prepare and copy
- mitosis — divide the nucleus
- cytokinesis — split the cell

## Watch the model

- Begin with one parent cell. During interphase it grows, increases subcellular structures and replicates its DNA so each chromosome has a copied partner.
- During mitosis, condensed chromosomes become visible, arrange and separate so one copy of each chromosome reaches each end of the cell.
- During cytokinesis, the cytoplasm and membrane divide. The outcome is two daughter cells, not one cell with twice the normal chromosome number.

## Which event belongs to interphase rather than mitosis?

- A. Copied chromosomes align across the middle of the cell.
- B. The cell replicates its DNA before nuclear division.
- C. Sister chromosome copies move to opposite ends of the cell.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.



## Build the explanation

- Sequence the evidence: prophase — chromosomes condense; metaphase — chromosomes align near the cell equator; anaphase — sister chromatids separate towards opposite poles.
- Complete the sequence: telophase — two chromosome groups reach the poles and new nuclei form; cytokinesis then divides the cytoplasm and membrane.
- Check the boundary: interphase precedes mitosis, while cytokinesis completes cell division. Stage boundaries in real cells are gradual rather than perfectly sharp.

## Find and repair the misconception

Cell-cycle glitch: DNA is copied during mitosis, then cytokinesis doubles the chromosome number in each daughter cell.

A. Keep it: every division should double chromosome number.

B. Repair it: DNA is copied in interphase, mitosis separates copies and cytokinesis forms daughters with the parent chromosome number.

C. Repair it: remove DNA copying because daughter cells do not need genetic information.

## Cell-cycle event routing console

Route each event to preparation before nuclear division, nuclear division, or cell split/outcome, then build one defensible sequence.

- Read one event card and identify what changes.
- Commit it to interphase, mitosis or cytokinesis/outcome.
- Place mitotic events in chromosome-evidence order.
- Freeze the sequence only when DNA copying and chromosome separation are not confused.

## Cell-cycle event routing console

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

## Choose a group · defend your placement

- INTERPHASE · PREPARE
- MITOSIS · NUCLEAR DIVISION
- CYTOKINESIS · CELL SPLIT

Describe the complete cycle using interphase, one visible clue for each mitotic stage, and cytokinesis.

## Your independent mission

Create an evidence-labelled cell-cycle sequence and answer GCSE DESCRIBE and EXPLAIN questions that distinguish interphase, mitosis and cytokinesis.

Record: A complete ordered cycle, at least four stage-evidence links, a correct distinction between DNA replication and chromosome separation, and one pupil-owned explanation.

## Choose your support and challenge

- Supported: Order six labelled picture cards, match prepare/divide/split, and complete a stem explaining when DNA is copied.
- Standard: Sequence interphase, four named mitotic stages and cytokinesis, annotate visible evidence, then explain why DNA replication must precede mitosis.
- Stretch: Evaluate an incorrect cell-cycle model, repair chromosome-number reasoning and explain why stage boundaries can be uncertain in a snapshot.

## Show what you know

Describe the cell cycle from interphase to two daughter cells using at least four formal stage terms.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.